Kolmogorov-Smirnov Test

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Introduction

The Kolmogorov-Smirnov (KS) test compares an empirical distribution function (ECDF) against either a hypothesised theoretical CDF (one-sample test) or another empirical distribution (two-sample test). Its test statistic is the supremum of the absolute difference between the two CDFs, a measure of the largest vertical gap on a CDF plot. KS is the prototypical CDF-based goodness-of-fit test, valued for being fully non-parametric and distribution-free in the one-sample case (the null distribution of D does not depend on the specific F_0 for continuous distributions). It is widely used as a quick distributional diagnostic and as a flexible two-sample test sensitive to differences in location, scale, and shape.

Prerequisites

A working understanding of empirical CDFs, the Glivenko-Cantelli theorem, and the difference between testing a fully specified hypothesised distribution and testing a parametric family with parameters estimated from the data.

Theory

For the **one-sample** test with sample $X_1, \ldots, X_n$ and hypothesised CDF F_0,

$$D = \sup_x |\hat{F}_n(x) - F_0(x)|;$$

under $H_0 : X \sim F_0$, $\sqrt{n} \cdot D$ has the Kolmogorov limiting distribution. For the **two-sample** test with empirical CDFs $\hat{F}_n$ and $\hat{G}_m$,

$$D_{n,m} = \sup_x |\hat{F}_n(x) - \hat{G}_m(x)|.$$

KS is distribution-free in the continuous one-sample case: the null distribution of D depends only on the sample size, not the specific F_0. When parameters of F_0 are estimated from the same data, the nominal p-values are anti-conservative; the Lilliefors correction provides the appropriate adjustment for testing Normality with estimated mean and SD.

Assumptions

The underlying distribution is continuous (ties produce warnings and approximate p-values), observations are independent and identically distributed, and parameters of F_0 are either fully specified or appropriately corrected for if estimated from the sample.

R Implementation

```r
set.seed(2026)

x <- rlnorm(100, meanlog = 0, sdlog = 0.8)
```

```r
ks.test(x, "pnorm", mean = mean(x), sd = sd(x))

z <- scale(x)
ks.test(z, "pnorm")

x1 <- rnorm(50, 0, 1)
x2 <- rnorm(60, 0.4, 1.2)
ks.test(x1, x2)

library(nortest)
lillie.test(x)
```

Output & Results

The script demonstrates the four standard use cases. The one-sample KS against a Normal with sample-estimated parameters rejects the null but the p-value is anti-conservative; the Lilliefors-corrected variant gives the proper rejection. The two-sample KS detects the location and scale shift between X_1 and X_2 ($D = 0.31$, $p = 0.007$), illustrating the test's sensitivity to general distributional differences.

Interpretation

A reporting sentence: "The Lilliefors-corrected Kolmogorov-Smirnov test rejected Normality of the residuals from the regression model ($D = 0.16$, $p < 0.001$), confirming the visual impression of right-skew in the Q-Q plot. A log transformation was applied before re-fitting; the residuals on the transformed scale passed the same test ($D = 0.05$, $p = 0.21$). The two-sample KS comparing transformed-residual distributions across treatment arms detected no significant difference in shape ($D = 0.08$, $p = 0.52$)." Always pair KS with a Q-Q plot.

Practical Tips

- KS is generally less powerful than Shapiro-Wilk for testing Normality in small to moderate samples; prefer Shapiro-Wilk as the default Normality test and use KS when the hypothesised distribution is non-Normal or when a two-sample distributional comparison is needed.
- Ties in the sample invalidate the strict KS p-value and R issues a warning; prefer Anderson-Darling or the Cramér-von Mises test with discretised or rounded data, which handle ties more gracefully.
- The Lilliefors correction must be used when testing Normality with sample-estimated mean and SD; the standard `ks.test()` against `pnorm` with sample estimates gives anti-conservative p-values that reject too rarely.
- The two-sample KS detects any distributional difference — location, scale, shape — and is therefore a useful omnibus diagnostic when the analyst is unsure which aspect of the distribution might differ; it is generally less powerful than the Mann-Whitney U test for pure location shifts.
- For very large samples, KS will reject almost any parametric fit because real data have rounding, ceiling effects, and other minor non-Normalities; use effect-size summaries and Q-Q plots in large samples rather than relying on the p-value alone.
- The Anderson-Darling test gives more weight to the tails of the distribution than KS does; for tail-sensitive applications (extreme-value detection, financial risk), Anderson-Darling is often preferred.

R Packages Used

Base R `ks.test()` for one- and two-sample Kolmogorov-Smirnov; `nortest::lillie.test()` for the Lilliefors-corrected Normality variant; `nortest::ad.test()` and `nortest::cvm.test()` for Anderson-Darling and Cramér-von Mises alternatives; `dgof::ks.test()` for discrete-data extensions; `car::qqPlot()` for graphical complements to the formal test.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests